

Fine-tuning multilingual pretrained NMT models for a morphologically rich low-resource language (Luganda–English)

Qin Wang, Wenyu Li, Zini Bie

Abstract

Although machine translation (MT) is well developed for high-resource languages, it remains underexplored for low-resource languages. This paper aims to bridge the research gap between high-resource and low-resource languages by focusing on Luganda–English translation. We fine-tuned two multilingual neural machine translation (NMT) models—No Language Left Behind (NLLB)-600M and mBART-50—on the Luganda–English language pair and compared their performance with four baselines: the two pretrained models, a copy baseline, and Google Translate. Our results show that the pretrained NLLB model already performs reasonably well, achieving a Bilingual Evaluation Understudy (BLEU) score of 15.62, whereas mBART-50 achieved a lower score of 7.60, likely because Luganda was not included in its pretraining data. Fine-tuning significantly improved the BLEU scores for both models. Moreover, we applied back-translation to further enhance the NLLB model; however, it did not lead to additional improvements. Overall, these findings demonstrate the effectiveness of fine-tuning for MT in low-resource languages and highlight potential challenges of applying back-translation when training data quality is limited.

1 Introduction

Machine translation for low-resource languages faces significant challenges, especially for morphologically rich languages like Luganda with limited training data. Our work focuses on Luganda to English translation, where morphological complexity and data scarcity limit translation quality.

We evaluate baseline multilingual models (mBART-50, NLLB-600M, Google Translate), then demonstrate the effectiveness of fine-tuning.

Our contribution is a systematic evaluation framework that shows fine-tuning models achieves great improvements. Specifically, NLLB-600M fine-tuning achieves strong performance compared to fine-tuned mBART-50.

2 Related Work

Neural machine translation has achieved remarkable success on high-resource languages, leaving low-resource languages still at a disadvantage (Kimera et al., 2022). Kisitu et al. (2019) highlighted that studying Luganda–English translation is beneficial for education in Uganda, since Luganda is widely used in local schools. This motivates us to conduct research on this low-resource language pair.

However, it is not easy to find suitable datasets for low-resource languages. Agić and Vulić (2019) introduced JW300, a parallel corpus covering more than 300 languages, including many low-resource ones. Later, Nabende et al. (2023) introduced Sunbird African Language Technology (SALT), a collection of parallel corpora and translation models for Ugandan languages, including Luganda–English pairs. Considering that the SALT data are more natural and cover more locally relevant topics, we used the Luganda–English pairs from SALT for our experiments.

In addition to data, finding suitable models is crucial. Team et al. (2022) proposed NLLB, a multilingual translation system aiming to reduce the performance gap between high-resource and low-resource languages. It is highly relevant to our research. Due to the limited computational resources, we used the compact 600M parameter version. Liu et al. (2020) proposed mBART, a multilingual sequence-to-sequence model pretrained on multiple languages. It performs well on low-resource translation tasks and can be fine-tuned for languages not included in its pretraining process.

Compared with previous work, our study focuses on fine-tuning and evaluating multilingual pretrained NMT models for the Luganda–English pair, as well as applying back-translation to further enhance performance. The fine-tuned models are benchmarked against Google Translate. The results show that our fine-tuned models achieve competitive BLEU scores, demonstrating the potential of transfer learning for this language pair.

3 Project Description

3.1 Data

In our project, we choose to use SALT corpus [Nabende et al. \(2023\)](#) which contains about 25,000 Luganda–English sentence pairs. The dataset is divided into a training set of 23,497 sentence pairs, a validation set of 496 sentence pairs, and a test set of 500 sentence pairs. The corpus provides sentence-aligned parallel text and covers diverse domains relevant to Uganda, including:

- Agriculture
- Health & sanitation
- Culture and social topics

Luganda is a Bantu language spoken by an estimated 7 million speakers in Uganda. And it shows several linguistic features that cause challenges for machine translation:

- Over 10 noun classes with complex agreement patterns
- Tonal patterns that affect word meaning
- Extensive prefix and suffix morphology

These features result in high surface variation, making translation particularly difficult where limited digital corpora and computational linguistics research are available.

3.2 Model and Experimental Setup

We evaluated four baseline translation systems and two multilingual pretrained models that were fine-tuned on Luganda–English data.

Baseline Models

1. Copy Baseline

Return the source Luganda input as output without translation, establishing the lower bound for translation performance.

2. mBART-50 Baseline

Model: mBART-50¹

Config: The model is configured with `src_lang="sw_KE"` and `tgt_lang="en_XX"`, which specify the source and target language identifiers used by mBART-50 for multilingual processing. Since Luganda is not a pretrained language, we use the similar Swahili language as the source language.

Process: The model applied tokenization to the input (maximum sequence length of 128; truncation enabled), followed by translation generation using beam search (beam size = 4). The output was obtained through decoding, and intermediate checkpoints are saved every 50 sentences.

3. NLLB-600M baseline

Model: NLLB-600M²

Config: The model used the language identifiers `source language="lug_Latn"` and `target language="eng_Latn"`, corresponding to Luganda and English written in Latin script. These language tags specify the tokenizer and decoder languages for translation.

Process: The pipeline mirrored that used for mBART-50. This included input tokenization, sequence generation, and decoding to produce the final translation output.

4. Google Translate

Translation requests were sent using the language identifiers `sl='lg'` (Luganda) and `tl='en'` (English). To comply with service rate limitations, a delay of 0.5 seconds was inserted between successive calls. Processing the full test set required approximately 20–30 minutes.

Fine-tuned Models

Two multilingual pretrained models were fine-tuned using approximately 24,000 sentence pairs. We also trained an NLLB-600M variant with back-translation to incorporate synthetic parallel data.

1. mBART-50

Originally, we planned to fine-tune the

¹<https://huggingface.co/facebook/mbart-large-50-many-to-many-mmt>

²<https://huggingface.co/facebook/nllb-200-distilled-600M>

mBART-50 model on the train dataset for three epochs. But the training process was stopped after 2 complete epochs due to disk quota limitations. Training logs showed consistent loss reduction (from roughly 2.68 to 1.05) and improved validation BLEU (from 22.50 to 44.60). We therefore used the last available checkpoint (after two epochs), which achieved a BLEU score of 25.93 on the test set, improving upon the baseline score of 7.60. The fine-tuning configuration for mBART-50 is presented in Figure 1.

Setting	Description
Model	facebook/mbart-large-50-many-to-many-mmt
Dataset	SALT parallel corpus (Luganda–English)
Training size	23,947 sentence pairs
Development size	496 sentence pairs
Testing size	500 sentence pairs
Batch size	8
Initial learning rate	2×10^{-5}
Fine-tuning epochs	planned 3 (terminated after ≈ 2 due to storage limit)
Hardware	Tesla T4 GPU
Time	≈ 20 hours on a single GPU before crashing

Figure 1: Fine-tuning configuration for mBART-50

2. NLLB-600M

We fine-tuned NLLB-600M using the same pipeline. After fine-tuning, performance improved from 15.62 to 35.99 BLEU, representing the best performance among all evaluated models. The fine-tuning configuration for NLLB-600M is presented in Figure 2.

Setting	Description
Model	facebook/nllb-200-distilled-600M
Dataset	SALT parallel corpus (Luganda–English)
Training size	23,947 sentence pairs
Development size	496 sentence pairs
Testing size	500 sentence pairs
Batch size	8
Initial learning rate	2×10^{-5}
Fine-tuning epochs	5
Hardware	Tesla T4 GPU
Time	≈ 2 hours on a single GPU

Figure 2: Fine-tuning configuration for NLLB-600M

3. Back-Translation

Furthermore, we fine-tuned an additional NLLB-600M variant using back-translation.

First, we fine-tuned a reverse-direction model (English→Luganda) using the same training and development data. Then, we used this model to translate the English side of the

training corpus into Luganda, creating synthetic parallel data. The synthetic data were combined with the original dataset to train another forward model (Luganda→English).

However, the resulting BLEU score (35.99) did not show improvement over the model trained without back-translation.

3.3 Evaluation metric

All models, including both baseline and fine-tuned versions, were evaluated on the same test set using identical generation parameters to ensure comparability. We used the BLEU score as the primary evaluation metric.

BLEU is a widely used automatic metric for evaluating machine translation quality. It measures the overlap between the machine-generated translation and one or more human reference translations by comparing n-gram precision while applying a brevity penalty to discourage overly short outputs.

Higher BLEU scores indicate closer matches to the reference translations and thus better translation quality. However, BLEU mainly captures surface-level similarity and may not fully reflect semantic adequacy or fluency. Therefore, qualitative error analysis was also conducted in addition to BLEU evaluation.

4 Results and Analysis

4.1 Results

The results of the four baseline systems, two fine-tuned models, and the fine-tuned model with back-translation are presented in Table 1 and Figure 3. Among all systems, the Fine-tuned NLLB-600M and Fine-tuned NLLB-600M + Back Translation models achieved the highest BLEU score of 35.99, indicating strong translation capability. As expected, the Copy Baseline performed the worst. Overall, fine-tuning substantially improved translation performance compared with the baseline systems.

For mBART, fine-tuning allowed the model to acquire task-specific knowledge of Luganda–English translation while retaining general multilingual knowledge from pretraining, leading to a BLEU improvement of 18.33. For NLLB-600M, the use of a larger set of Luganda–English pairs further enhanced its ability to capture translation patterns, improving its BLEU score by 20.37.

However, back-translation did not bring additional improvements over the fine-tuned NLLB-600M. This might be due to the relatively low quality of the original training data, which could have been degraded further by synthetic data from the back-translation process. Another possible reason is that the fine-tuning hyperparameters were not sufficiently adjusted for the augmented dataset.

Interestingly, both fine-tuned models outperformed Google Translate. Despite its broad language coverage, Google Translate may suffer from limited and lower-quality training data for low-resource languages such as Luganda.

Model	BLEU
Copy Baseline	2.84
Google Translate	9.24
mBART-50 Baseline	7.60
NLLB-600M Baseline	15.62
Fine-tuned mBART-50	25.93
Fine-tuned NLLB-600M	35.99
Fine-tuned NLLB-600M + Back translation	35.99

Table 1: Model performance on the SALT Luganda–English dataset.

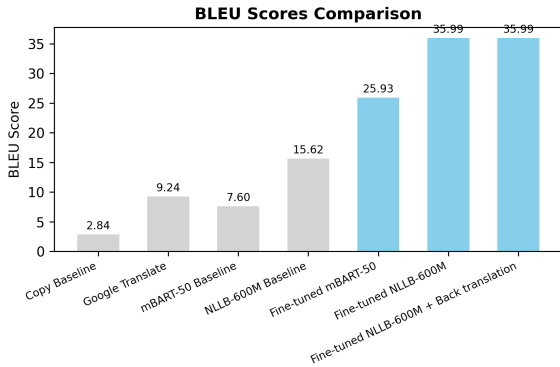


Figure 3: BLEU score comparison.

4.2 Analysis

We conducted an error analysis on three representative sentences to better understand the translation behaviour of different models. Tables 2–4 present the translations and analyses for three examples.

The first sentence is "Emikolo kitundu ku bu-lamu.", the second is "Ebikoola bya kasooli biri-iriddwa ebisaanyi." and the third is "Ekikoola kya

kasooli kya kyenvu wabula langi yaakyo etera okuba eya kitaka wansi."

The results are consistent with the BLEU scores reported in the previous section. After fine-tuning, both the mBART-50 and NLLB-600M models showed substantial improvements in translation quality, while the back-translation approach did not lead to further gains, which aligns with the quantitative results.

Specifically, the fine-tuned models produced more semantically accurate translations and captured domain-specific agricultural terms correctly, whereas the baseline models often generated unrelated or meaningless sentences.

Among all systems, Google Translate performed well on the first sentence, but for the second and the third sentences, it produced only partial structural alignment without conveying the correct meaning or word order.

5 Discussion

This project investigates machine translation for Luganda, a morphologically rich and low-resource African language. Using the SALT corpus, we evaluate several baselines, fine-tune two multilingual pretrained models, and try to use back-translation to improve the results.

Our experiments reveal that while baseline scores are relatively low (mBART-50: 7.6 BLEU; NLLB-600M: 15.62 BLEU), fine-tuning makes substantial improvements:

- mBART-50: 7.6 $\rightarrow$ 25.9 BLEU
- NLLB-600M: 15.62 $\rightarrow$ 35.99 BLEU

Both fine-tuned models outperform Google Translate (9.24 BLEU) by a large margin, demonstrating the effectiveness of transfer learning techniques in low-resource machine translation.

We also experimented with an augmented variant of NLLB-600M incorporating back-translation. However, the BLEU score (35.99) is the same when not using this strategy, suggesting that the benefits of synthetic data may depend on data quality and correct fine-tuning hyperparameters.

Despite limitations in dataset size and computational resources, our study highlights:

- Multilingual pretrained models are highly effective for Luganda translation.

Model	Translation	Analysis
Reference	Ceremonies are part of life.	Reference translation.
mBART-50 Baseline	I'm like, I'm in the middle of nowhere.	Wrong meaning; unrelated content
NLLB-600M Baseline	Schools are a part of life.	Wrong meaning; partial structural alignment.
Fine-tuned mBART-50	Celebrations are part of life.	Meaning mostly correct; minor lexical difference.
Fine-tuned NLLB-600M	Ceremonies are part of life.	Correct translation.
Fine-tuned NLLB-600M + Back Translation	Ceremonies are part of life.	Correct translation.
Google Translate	Ceremonies are part of life.	Correct translation.

Table 2: Error Analysis of Sentence 1 — Model Outputs vs Reference

Model	Translation	Analysis
Reference	The maize leaves have been eaten by caterpillars.	Reference translation.
mBART-50 Baseline	The Ebiscoola was one of only two species of bison.	Wrong meaning; unrelated content; one untranslated word.
NLLB-600M Baseline	The spoons of the capsules are cooked in a saucepan.	Wrong meaning; unrelated content.
Fine-tuned mBART-50	The maize leaves have been cut long.	Incorrect meaning; partial grammatical alignment; correct agricultural term “maize leaves.”
Fine-tuned NLLB-600M	The maize leaves have been eaten raw.	Same as Fine-tuned mBART-50
Fine-tuned NLLB-600M + Back Translation	The maize leaves have been eaten raw.	Same as Fine-tuned NLLB-600M
Google Translate	Corn leaves are eaten by weeds.	Wrong tense and meaning; partial grammatical alignment.

Table 3: Error Analysis of Sentence 2 — Model Outputs vs Reference

Model	Translation	Analysis
Reference	The maize leaf is yellow; however, its colour tends to brown at the base.	Reference translation.
mBART-50 Baseline	The equation for the meaning of the four-pointed wavelength of a star may have been derived from the fact that it was a single wavelength.	Completely wrong meaning; unrelated content.
NLLB-600M Baseline	The capsule’s skin is gray, but its color is usually brownish on the underside.	Wrong meaning; partial structural alignment.
Fine-tuned mBART-50	The yellow bark of a maize leaf usually has a lower bark.	Wrong meaning; right agricultural vocabulary- (“maize leaf”).
Fine-tuned NLLB-600M	The maize leaf is yellow but its color is usually brown below.	Correct meaning and grammatical structure; right agricultural vocabulary- (“maize leaf”).
Fine-tuned NLLB-600M + Back Translation	The maize leaf is yellow but its color is usually brown below.	Same as fine-tuned NLLB; consistent but no improvement.
Google Translate	The corn root is yellow but its color is often brown down.	Wrong meaning; wrong word order; partial alignment in structure.

Table 4: Error Analysis of Sentence 3 — Model Outputs vs Reference

- Fine-tuning significantly improves performance, even with limited data.

Based on our findings, future work should explore multiple directions to further improve translation quality and model generalization:

- **Data augmentation and synthetic corpora:** Expand the dataset using large-scale back-translation, round-trip translation, or data mining from web sources to increase domain and lexical coverage.
- **Scaling model size and capacity:** Investigate the performance of larger NLLB variants (1.3B and 3.3B parameters) or other multilingual architectures to assess scaling effects on low-resource learning.
- **Evaluation beyond BLEU:** Include human evaluation and metrics such as COMET or ChrF to capture semantic adequacy and fluency more accurately.

6 Conclusion

In summary, our findings confirm that fine-tuning multilingual pretrained models can significantly enhance translation performance for low-resource languages. By scaling models, enriching data,

and improving evaluation, future work can further enhance translation quality for low-resource languages such as Luganda.

References

- Željko Agić and Ivan Vulić. 2019. Jw300: A wide-coverage parallel corpus for low-resource languages. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019)*, pages 3204–3210.
- Richard Kimera, Daniela N. Rim, and Heeyoul Choi. 2022. [Building a parallel corpus and training translation models between luganda and english](#). *Journal of KIISE*, 49(11):1009–1016. Also available at arXiv:2301.02773 [cs.CL].
- Ronald Kisitu, Monica Nabirye, and Guy De Pauw. 2019. English to luganda smt: Ganda noun class prefix segmentation for enriched machine translation. In *Proceedings of the 2nd Workshop on Technologies for MT of Low Resource Languages (LoResMT 2019)*, Dublin, Ireland.
- Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike

Lewis, and Luke Zettlemoyer. 2020. [Multilingual denoising pre-training for neural machine translation](#). *Transactions of the Association for Computational Linguistics*, 8:726–742.

Patrick Nabende, Monica Nabirye, Syed Jameel, and Janet Nabukenya. 2023. Machine translation for african languages: Community creation of datasets and models in uganda. In *Proceedings of the 4th Workshop on African Natural Language Processing (AfriNLP 2023)*, Dubrovnik, Croatia.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Sermarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. [No language left behind: Scaling human-centered machine translation](#). ArXiv preprint arXiv:2207.04672.